

Anti-inflammatory effects of a low arachidonic acid diet and fish oil in patients with rheumatoid arthritis.

BACKGROUND: Patients with rheumatoid arthritis (RA) improve on a vegetarian diet or supplementation with fish oil. We investigated the effects of both dietary measures, alone and in combination, on inflammation, fatty acid composition of erythrocyte lipids, eicosanoids, and cytokine biosynthesis in patients with RA. **METHODS:** Sixty-eight patients with definitive RA were matched into two groups of 34 subjects each. One group was observed for 8 months on a normal western diet (WD) and the other on an anti-inflammatory diet (AID) providing an arachidonic acid intake of less than 90 mg/day. Patients in both groups were allocated to receive placebo or fish oil capsules (30 mg/kg body weight) for 3 months in a double-blind crossover study with a 2-month washout period between treatments. Clinical examination and routine laboratory findings were evaluated every month, and erythrocyte fatty acids, eicosanoids, and cytokines were evaluated before and after each 3-month experimental period. **RESULTS:** Sixty patients completed the study. In AID patients, but not in WD patients, the numbers of tender and swollen joints decreased by 14% during placebo treatment. In AID patients, as compared to WD patients, fish oil led to a significant reduction in the numbers of tender (28% vs 11%) and swollen (34% vs 22%) joints ($P<0.01$). Compared to baseline levels, higher enrichment of eicosapentaenoic acid in erythrocyte lipids (244% vs 217%) and lower formation of leukotriene B(4) (34% vs 8%, $P>0.01$), 11-dehydro-thromboxane B(2) (15% vs 10%, $P<0.05$), and prostaglandin metabolites (21% vs 16%, $P<0.003$) were found in AID patients, especially when fish oil was given during months 6-8 of the experiment. **CONCLUSION:** A diet low in arachidonic acid ameliorates clinical signs of inflammation in patients with RA and augments the beneficial effect of fish oil supplementation.